

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} d(\beta) &= (a_1T_1 + a_2T_2)^2 + (b_1T_1 + b_2T_2)^2 + (a_1T_1 + a_2T_2)(b_1T_1 + b_2T_2) + (a_3^2 + a_3b_3 + b_3^2)T_3^2, \\ d(\gamma) &= (a_1T_1 + a_2T_2)(b_1T_1 + b_2T_2)^2 + (a_1T_1 + a_2T_2)^2(b_1T_1 + b_2T_2) + (a_3^2b_3 + a_3b_3^2)T_3^3. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} L_{\omega, \mathbf{c}}(V) &= \sum_{j=1}^3 \left(C_{1,j} \|\nabla v_j\|_{L^2}^2 + C_{2,j} \|v_j\|_{L^2}^2 + \sum_{k=1}^d C_{3,j,k} \|\partial_k v_j + c_{j,k} v_j\|_{L^2}^2 \right) \\ &= \sum_{j=1}^3 \left(C_{1,j} \sum_{k=1}^d \|\partial_k v_j\|_{L^2}^2 + C_{2,j} \|v_j\|_{L^2}^2 + \sum_{k=1}^d C_{3,j,k} \|\partial_k v_j + c_{j,k} v_j\|_{L^2}^2 \right) \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \rho(M, M') &:= \lim_{\Delta t \rightarrow 0^+} \frac{1}{\Delta t} \mathbb{P}(Q(t + \Delta t) = M' \mid Q(t) = M) \\ &= \begin{cases} \frac{\pi_{ij}(x_i - m_{i,\cdot})(y_j - m_{\cdot,j})}{n - m_{tot}} & \text{if } M' = M + I^{ij}, \text{ and} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{2\sqrt{6}} \sqrt{3 - \frac{3}{\sqrt{5}}} \begin{pmatrix} \omega + 1 \\ i \\ \omega - 1 \\ i \end{pmatrix} + \frac{1}{2\sqrt{2}} \sqrt{1 + \frac{3}{\sqrt{5}}} \begin{pmatrix} 0 \\ \omega \\ 0 \\ -\omega \end{pmatrix}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \Omega_1 &= -\frac{1}{4}(F_2 + F_3 + F'_4) + \cdots, & \Omega_2 &= -\frac{1}{4}(-F_1 + F_3 + F'_4) + \cdots, \\ \Omega_3 &= -\frac{1}{4}(-F_1 - F_2 + F'_4) + \cdots, & \Omega_4 &= -\frac{1}{4}(-F_1 - F_2 - F_3) + \cdots. \end{aligned}$$